MATH 243 — Probability and Statistics

Tests of Hypothesis: z-test, t-test, Two-sample and Paired Tests, Proportions, Correlation, χ²

Lesson notes · Exam approach · TF practice with solutions · Quick notes (KKS Sir's portion — Probability & ANOVA)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Quick Notes — Tests of Hypothesis — revision bullets

NOTE Page numbers refer to KKS Sir's Lecture-13-20- Hypothesis.pdf (44 pp.), written as Hyp p. n. This lecture was added after the first version of these notes; it now covers the inference questions that were previously listed under Outside Notes. Goodness-of-fit χ² and the unpooled Welch degrees of freedom are only sketched in the lecture and are completed from the textbook (marked). Critical values are from the t, F, χ² and z tables in the course folder (or as printed on the paper). "TF p." = merged question PDF page; "22 batch p." = 2023-24 paper.

Exam Map — What the TF Papers Ask

Session (paper)	Questions answered by this note [lesson]
2023-24 (MATH 243) TF p. 22 batch p.3, 22 batch p.4	Q6(c) Stratified random sampling (example); delivery time claim < 120 min, σ 30, n 49, mean 100 (z = −1.645) (12) [L21] · Q7(a) Short notes: goodness of fit, one-/two-tailed tests, degrees of freedom, null/alternative hypothesis (10) [L20] · Q7(b) Machine: 16/500 imperfect before, 3/100 after overhaul — improved? (1 %) (12) [L23] · Q7(c) Two samples of 8 and 7 weights: is the difference of means significant? (t13 = 2.16) (13) [L22] · Q8(a) Education vs job performance 3×3 table, 200 employees: χ² test ($\chi^2_{0.05,4}$ = 9.49) (15) [L24]
2022-23 (MATH 243) TF p. 50	Q7(a) Differentiate: statistic/parameter, critical/acceptance region, exact/inexact hypothesis, uni/non-directional test, level of significance/p-value (10) [L20] · Q7(b) GPA: n = 40, mean 2.8, sd 0.35 vs 2.4; 99 % CI (12) [L21] · Q7(c) Bank waiting time: n = 12, mean 42, sd 11.9 — is μ < 50? (t11 = 1.796) (13) [L21] · Q8(a) Drug vs typhoid 2×2 table: χ² test of effectiveness (χ^2_1 = 3.84) (10) [L24]
2021-22 (MATH 245) TF p. 45	Q8(a) Heights of 10 individuals: is the mean height 65? (t9 = 2.262) (17) [L21]
2020-21 (MATH 245) TF p. 39	Q8(b) Cereal sodium: n = 20, mean 244, sd 24.5 — is μ > 220? (5 %) (15) [L21] · Q8(c) DNA paternity testing: describe Type I and Type II errors (10) [L20]
2019-20 (MATH 245) TF p. 34	Q6(b) Mice diet: 64 mice, mean 38, sd 5.8 — is μ < 40? (P-value) (18) [L21]
2018-19 (MATH 245) TF p. 25	Q8(b) GPA: n = 40, mean 2.8, sd 0.35 vs population 2.4 (1 %); 99 % confidence interval (17) [L21]
2016-17 (MATH 245) TF p. 21	Q8(b) 100 deaths, mean life 61.8, σ 4.9: is the mean life span greater than 60? (5 %) (15) [L21]
2015-16 (MATH 241) TF p. 16	Q8(c) Chocolate sales: 153.7 before; 29 stores after, mean 169.4, sd 19.7 — was advertising successful? (5 %) (10) [L21]
2014-15 (MATH 241) TF p. 11	Q7(b) Bulbs: claimed mean 4 years, σ 0.16; sample of 10 has mean 3.45 — is the claim justified? (5 %) (15) [L21]
2013-14 (MATH 241) TF p. 8	Q8(a) 11 tins of talcum powder: can the average packing be 0.5 lb? (t-test, 5 %) (20) [L21] · Q8(b) 320 families with 5 children: χ² test of equal male/female births (15) [L24]
2012-13 (MATH 241) TF p. 5	Q7(b) 640 families with 5 children: χ² test that male and female births are equally probable (17) [L24] · Q8(b) Two supermarkets, 500 shoppers each (Tk 500, sd 80 vs Tk 440, sd 90): are mean expenditures equal? (10 %) (17) [L22]
2011-12 (MATH 241) TF p. 3	Q8(a) Type I / Type II errors; t-test: 9 items, can the population mean be 47? (5%) (17) [L21] · Q8(b) 5 coins tossed 3200 times (heads 0…5): χ² test that the coins are unbiased (18) [L24]

EXAM FOCUS 23 questions, 328 marks (13 questions and 167 marks since 2019-20). Every paper from 2011-12 on has one 10–20 mark test: a one-sample z or t test (most often), a χ² test (2011-12, 2012-13, 2013-14, 2022-23, 2023-24), a two-sample test or a two-proportion test (2023-24). Learn the 5-step layout — hypotheses, α, test statistic, critical region, decision — and choose the test by the three questions: is σ known? is n large (≥ 30)? one sample or two?

Lesson 20 — Concepts: Hypotheses, Errors, Level of Significance, p-value

Hyp pp. 1–6, 11–12

1. Hypothesis, Null and Alternative (pp. 1–2)

A **statistical hypothesis** is a statement or assumption about a parameter, written H. The **null hypothesis** H_0 is the hypothesis stated for possible rejection, assumed true until the data show otherwise; the **alternative hypothesis** H_1 is the contrary statement. For a sample from f(x, θ): H_0: $\theta = \theta_0$ versus H_1: $\theta \neq \theta_0$, where θ_0 is the specified (null) value.

2. Critical (Rejection) Region and Tails (pp. 2–4)

Split the sample space S into w and S − w. If the sample point falls in w, reject H_0 — w is the **critical (rejection) region**; S − w is the **acceptance region**.

- **Two-tailed** test: critical region at both ends — H_1: $\mu \neq \mu_0$.
- **One-tailed** test: critical region at one end — right tail H_1: $\mu > \mu_0$, left tail H_1: $\mu < \mu_0$.
- Put the claim you need evidence for in H_1 (e.g. "proportion defective is at most 3 %": H_0: $p = 0.03$, H_1: $p < 0.03$; "users increased": H_0: $\mu_2 - \mu_1 = 0$, H_1: $\mu_2 - \mu_1 > 0$).

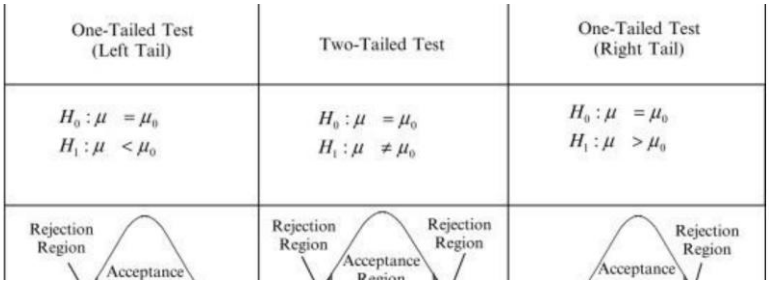


Figure 20.1 — Acceptance and rejection regions of one-tailed and two-tailed tests (Hyp p. 3)

3. Type I and Type II Errors, Level of Significance (p. 4)

Decision	H_0 true	H_0 false
Accept H_0	no error	Type II error (β)
Reject H_0	Type I error (α)	no error

α = P(reject H_0 | H_0 is true) = level of significance; β = P(accept H_0 | H_0 is false)

A Type I error (convicting an innocent defendant, operating on a patient who does not need surgery) is usually the more serious, so tests are designed with P(Type I) ≤ a small pre-assigned α (0.10, 0.05, 0.01); then the Type II error is made as small as possible. A low α needs a lot of evidence to reject H_0.

EXAM-READY ANSWER DNA paternity (2020-21 Q8(c)). H_0: the man is the father; H_a: he is not. **Type I error**: rejecting H_0 when it is true — concluding "not the father" for the real father. Since the test is 99.9 % accurate when he is the father, this happens with probability about 0.001 (a false exclusion). **Type II error**: accepting H_0 when it is false — concluding "is the father" for a man who is not. The test is 100 % accurate in that case, so this error has probability 0 (a false match cannot occur).

4. Test Statistic and the Four Tests (pp. 5–7)

Test statistic = (observed value − expected value under H_0) / standard error

Test	Statistic	Use
Normal (z)	$z = (u - E(u))/\sigma(u) \sim N(0, 1)$	σ known, or estimated from a large sample ($n \geq 30$)
t	$t = (u - E(u))/\hat{\sigma}(u) \sim t_v$	σ unknown, small sample ($n < 30$), normal population
χ^2	$\Sigma (O - E)^2/E \sim \chi^2_v$	distribution as a whole; independence of attributes
F	$F = S_1^2/S_2^2 \sim F_{v1, v2}$	equality of two variances (upper tail only) — ANOVA note

5. p-value (pp. 11–12)

The **p-value** is the probability, assuming H_0 is true, of a test statistic as extreme as, or more extreme than, the one observed. Decision: $p \leq \alpha \rightarrow$ reject H_0; $p > \alpha \rightarrow$ do not reject.

right tail: $p = P(Z \geq z_{cal}) = 1 - \Phi(z_{cal})$; left tail: $p = P(Z \leq z_{cal})$
two-tailed: $p = 2\,P(Z \geq |z_{cal}|) = 2[1 - \Phi(|z_{cal}|)]$

Bulb example: $z_{cal} = -3.35 \rightarrow p = 2(1 - 0.9996) = 0.0008 < 0.05 \rightarrow$ reject.

EXAM-READY ANSWER Short notes (2022-23 Q7(a), 2023-24 Q7(a)). **Statistic vs parameter**: a statistic is a numerical summary of a sample ($\bar{x}$, s); a parameter is a numerical characteristic of the population (μ, σ). **Critical vs acceptance region**: the set of sample values for which H_0 is rejected, and its complement. **Exact vs inexact hypothesis**: an exact (simple) hypothesis fixes the parameter completely ($\mu = 50$); an inexact (composite) one specifies a range ($\mu > 50$, $\mu \neq 50$). **Unidirectional vs non-directional test**: one-tailed (H_1: $\mu > \mu_0$ or $\mu < \mu_0$; rejection region in one tail) versus two-tailed (H_1: $\mu \neq \mu_0$; both tails, α/2 each). **Level of significance vs p-value**: α is the pre-chosen maximum probability of a Type I error; the p-value is computed from the data — the smallest α at which H_0 would be rejected. **Goodness of fit**: a χ^2 test of whether observed frequencies agree with those expected from a specified distribution. **Degrees of freedom**: the number of independent values free to vary in a statistic ($n - 1$ for the t-test of a mean, $(r - 1)(p - 1)$ for a contingency table, $k - 1 - m$ for goodness of fit with m estimated parameters).

Exam Approach

1. Define the parameter and write H_0 (with =) and H_1; the direction of H_1 sets the tail.
2. Fix α; choose the test (z or t, one or two samples) and its critical value(s).
3. Compute the test statistic; compare with the critical value (or compute p).
4. Decide, then state the conclusion in the words of the problem.

Practice — Lesson 20

(a) Actual — 2020-21 Q8(c) (TF p. 39). (b) Actual — 2022-23 Q7(a) (TF p. 50). (c) Actual — 2023-24 Q7(a) (22 batch p. 3).

(a) In a paper on DNA paternity testing: "The test is 100 % accurate if the man is not the father and 99.9 % accurate if he is." Consider H_0: a particular man is the father against H_a: he is not. Describe Type I and Type II errors in this context. (10)

(b) Differentiate: (i) statistic and parameter, (ii) critical region and acceptance region, (iii) exact and inexact hypothesis, (iv) unidirectional and non-directional test, (v) level of significance and p-value. (10)

(c) Write short notes on goodness of fit, one-tailed and two-tailed tests, degrees of freedom, null and alternative hypothesis. (10)

✓ Solution

(a)–(c) Model answers are the EXAM-READY ANSWER boxes of §3 and §5; for (c) add the goodness-of-fit paragraph of Lesson 24 §4 and the tail figure of §2.

Lesson 21 — One-sample Tests of a Mean (z and t) and Confidence Intervals

Hyp pp. 7–16

1. σ Known, or Estimated from a Large Sample (n ≥ 30): z-test (pp. 7–13)

Assumptions: random sample from a normal population (or n large); σ known, or s from a large sample.

$$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n}) \quad [\sigma \text{ known}], \quad z = (\bar{x} - \mu_0)/(s/\sqrt{n}) \quad [\sigma \text{ estimated, } n \geq 30]$$

$$s^2 = [\Sigma x^2 - n\,\bar{x}^2]/(n-1)$$

H_1	Reject H_0 if ($\alpha = 0.05$)	Reject H_0 if ($\alpha = 0.01$)
$\mu > \mu_0$	$z > 1.645$ (1.64)	$z > 2.33$
$\mu < \mu_0$	$z < -1.645$	$z < -2.33$
$\mu \neq \mu_0$	$\lvert z\rvert > 1.96$	$\lvert z\rvert > 2.58$

$$100(1-\alpha)\%\ \text{CI for } \mu: \ \bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n} \quad (95\ \%: \bar{x} \pm 1.96\sigma/\sqrt{n};\ 99\ \%: \bar{x} \pm 2.58\sigma/\sqrt{n})$$

Example 1 (pp. 9–10): bulbs claimed $\mu = 3$ years, $\sigma^2 = 0.5$, $n = 35$, $\bar{x} = 2.6$: $z = (2.6 - 3)/(0.7071/\sqrt{35}) = -3.347 < -1.96 \rightarrow$ reject; 95 % CI = $2.6 \pm 1.96(0.1195) = (2.366, 2.834)$. Example 2 (p. 12): IQ of 64 women, $\bar{x} = 115.5$, $s = 20$, $H_1: \mu > 110$, $\alpha = 0.01$: $z = 5.5/2.5 = 2.2 < 2.33 \rightarrow$ do not reject.

2. σ Unknown, Small Sample (n < 30): t-test (pp. 14–16)

$$t = (\bar{x} - \mu_0)/(s/\sqrt{n}) \sim t_{n-1}; \quad \text{CI: } \bar{x} \pm t_{\alpha/2,\,n-1}\,s/\sqrt{n}$$

$$\text{two-tailed: reject if } |t| > t_{\alpha/2,\,n-1}; \ \text{right: } t > t_{\alpha,\,n-1}; \ \text{left: } t < -t_{\alpha,\,n-1}$$

Example (p. 15): 10 workers' production [40, 35, 48, 43, 35, 39, 44, 37, 42, 38], claimed $\mu = 20$: $\bar{x} = 40.1$, $s = 4.175$, $t = 15.223 > t_{0.025,\,9} = 2.262 \rightarrow$ reject; the claim is false.

TRAP The test statistic uses s with divisor n − 1 (not n). If n ≥ 30 (or σ is given), use z; the paper's "given t" or "necessary chart" tells you which table it expects. State the d.f. whenever you use t.

Lecture exercises — answers: (1) steel wire, $\sigma = 150$, $n = 25$, $\bar{x} = 1312$, $H_1: \mu > 1250$, $\alpha = 0.01$: $z = 62/30 = 2.07 < 2.33 \rightarrow$ do not reject. (2) scrap, $\sigma = 1.6$, $n = 10$, $\bar{x} = 23.9$, $H_1: \mu < 25.5$: $z = -1.6/0.506 = -3.16 < -2.33 \rightarrow$ reject (modification works). (3) users, $\sigma = 800$, $n = 100$, $\bar{x} = 5200$: $z = 2.5 > 1.645 \rightarrow$ reject (increased). (4) steel wire, $n = 25$, $\bar{x} = 110$, $s^2 = 144$, $\mu_0 = 120$: $t = -10/(12/5) = -4.17$, $|t| > t_{0.025,\,24} = 2.064 \rightarrow$ reject.

Exam Approach

1. Parameter, H_0, H_1, α (as given).
2. Test: z if σ is known or n ≥ 30; t (n − 1 d.f.) if σ is unknown and n < 30.
3. Critical region for the tail of H_1; compute the statistic with all steps shown.
4. Decision and a conclusion in context; add the CI if asked (use the same table value as the test).

Practice — Lesson 21

(a) Actual — 2019-20 Q6(b) (TF p. 34). (b) Actual — 2016-17 Q8(b) (TF p. 21). (c) Actual — 2014-15 Q7(b) (TF p. 11). (d) Actual — 2018-19 Q8(b) = 2022-23 Q7(b) (TF pp. 25, 50). (e) Actual — 2020-21 Q8(b) (TF p. 39). (f) Actual — 2015-16

Q8(c) (TF p. 16). (g) Actual — 2022-23 Q7(c) (TF p. 50). (h) Actual — 2023-24 Q6(c) (22 batch p. 3). (i) Actual — 2021-22 Q8(a) (TF p. 45). (j) Actual — 2013-14 Q8(a) (TF p. 8). (k) Actual — 2011-12 Q8(a) (TF p. 3).

(a) A student claims that mice on a vitamin- and protein-enriched diet live about 40 months. 64 such mice have a mean life of 38 months with s = 5.8. Is there reason to believe $\mu < 40$? Use a P-value in your conclusion. (18)

(b) A random sample of 100 recorded deaths in Bangladesh showed an average life span of 61.8 years. Assuming $\sigma = 4.9$ years, does this indicate that the mean life span is greater than 60? Use $\alpha = 0.05$. (15)

(c) A bulb manufacturer claims an average life of 4 years with $\sigma = 0.16$ years. A random sample of 10 bulbs has a mean of 3.45 years. Does the sample justify the claim? Use $\alpha = 5\ \%$. (15)

(d) The mean and SD of GPA scores of a random sample of 40 students were 2.8 and 0.35. Can we conclude that the sample comes from a group with mean 2.4? Use $\alpha = 1\ \%$ ($z = \pm 2.58$) and compute the 99 % confidence interval for the population mean. (17 / 12)

(e) A random sample of 20 servings of a cereal has mean sodium content 244 mg and SD 24.5 mg. At $\alpha = 0.05$, is the average sodium content greater than 220 mg? (15)

(f) Mean weekly sale was 153.7 bars per store; after an advertising campaign the mean in 29 stores was 169.4 with SD 19.7. Was the campaign successful? $\alpha = 5\ \%$. (10)

(g) The average waiting time to cash a cheque has been 50 minutes. A new procedure: 12 customers, mean 42 minutes, SD 11.9. Test, at 5 %, that the mean is now less than 50 (normal; $t_{0.05} = 1.796$ for $v = 11$). (13)

(h) An online medicine shop claims a mean delivery time below 120 minutes with $\sigma = 30$. 49 orders have mean 100 minutes. Test at 0.05 ($z = -1.645$). (12)

(i) Heights (inches) of ten individuals: 63, 63, 64, 65, 66, 69, 69, 70, 70, 71. Discuss the suggestion that the mean height of the universe is 65 ($t_{9,\,0.05} = 2.262$). (17)

(j) 11 tins of talcum powder weigh (lb) 0.44, 0.51, 0.49, 0.52, 0.45, 0.48, 0.46, 0.45, 0.47, 0.45, 0.47. Can the average packing be taken as 0.5 lb at 5 % ($t = \pm 2.23$)? (20)

(k) Explain Type I and Type II errors. Nine items had values 50, 48, 49, 53, 52, 47, 45, 46, 51: can this sample be regarded as from a population with mean 47 at the 5 % level (given for $v = 8$, $t = 3.355$)? (17)

✓ Solution

(a)

```
H0: μ = 40   H1: μ < 40   α = 0.05
Test statistic: n = 64 ≥ 30 → z = (38 − 40)/(5.8/√64) = −2/0.725 = −2.759
Critical region: z < −1.645
Decision: p-value = P(Z ≤ −2.76) = 0.0029 < 0.05 → reject H0 (z = −2.76 < −1.645)
```

Conclusion: there is significant evidence that the mean life is less than 40 months; the claim of 40 months is not supported (p = 0.0029, significant even at 1 %).

(b)

```
H0: μ = 60   H1: μ > 60   α = 0.05
Test statistic: z = (61.8 − 60)/(4.9/√100) = 1.8/0.49 = 3.673
Critical region: z > 1.645
Decision: 3.67 > 1.645 → reject H0
```

Conclusion: the mean life span today is greater than 60 years.

(c)

```
H0: μ = 4   H1: μ ≠ 4   α = 0.05
Test statistic: σ = 0.16 known → z = (3.45 − 4)/(0.16/√10) = −0.55/0.0506 = −10.87
Critical region: |z| > 1.96
Decision: |−10.87| > 1.96 → reject H0
```

Conclusion: the sample mean does not justify the claim; mean longevity differs from 4 years (it is less). (If the paper's chart is the t-table, use t = −10.87 against $t_{0.025,\,9}$ = 2.262 — same decision.)

(d) σ estimated from n = 40 ≥ 30 → z-test.

```
H0: μ = 2.4   H1: μ ≠ 2.4   α = 0.01, critical region |z| > 2.58
z = (2.8 − 2.4)/(0.35/√40) = 0.4/0.0553 = 7.23   → |z| = 7.23 > 2.58 → reject H0
99% CI: 2.8 ± 2.58 × 0.0553 = 2.8 ± 0.143 = (2.657, 2.943)
```

The sample does not come from a group with mean 2.4; the mean score lies between 2.66 and 2.94 with 99 % confidence (2.4 is outside).

(e)

```
H0: μ = 220   H1: μ > 220   α = 0.05
Test statistic: n = 20 < 30, σ unknown → t = (244 − 220)/(24.5/√20) = 24/5.478 = 4.381  (19
d.f.)
Critical region: t > t_{0.05, 19} = 1.729
Decision: 4.38 > 1.729 → reject H0
```

Conclusion: the average sodium content per serving is significantly greater than 220 mg.

(f)

```
H0: μ = 153.7   H1: μ > 153.7   α = 0.05
Test statistic: n = 29 < 30 → t = (169.4 − 153.7)/(19.7/√29) = 15.7/3.658 = 4.292  (28 d.f.)
Critical region: t > t_{0.05, 28} = 1.701
Decision: 4.29 > 1.701 → reject H0
```

Conclusion: the advertising campaign was successful (sales rose significantly).

(g)

```
H0: μ = 50   H1: μ < 50   α = 0.05
Test statistic: t = (42 − 50)/(11.9/√12) = −8/3.435 = −2.329  (11 d.f.)
Critical region: t < −1.796
Decision: −2.33 < −1.796 → reject H0
```

Conclusion: the new procedure significantly reduces the mean waiting time below 50 minutes.

(h)

```
H0: μ = 120   H1: μ < 120   α = 0.05
Test statistic: z = (100 − 120)/(30/√49) = −20/4.286 = −4.667
Critical region: z < −1.645
Decision: −4.67 < −1.645 → reject H0
```

Conclusion: there is enough evidence to support the claim that the mean delivery time is less than 120 minutes.

(i)

```
x̄ = 67, s = 3.127, n = 10;  H0: μ = 65, H1: μ ≠ 65, α = 0.05
t = (67 − 65)/(3.127/√10) = 2.023   critical |t| > 2.262 (9 d.f.)
|2.02| = 2.02 < 2.262 → do not reject H0
```

The data are consistent with a universe mean height of 65 inches.

(j)

```
x̄ = 0.4718, s = 0.026, n = 11;  H0: μ = 0.5, H1: μ ≠ 0.5, α = 0.05
t = (0.4718 − 0.5)/(0.026/√11) = −3.594   critical |t| > 2.23 (10 d.f.)
|−3.59| > 2.23 → reject H0
```

The average packing cannot be taken as 0.5 lb (the tins are underfilled, $\bar{x}$ = 0.4773 lb).

(k) Type I and II errors: §3 of Lesson 20 (in this problem a Type I error would be wrongly concluding that the population mean is not 47).

```
x̄ = 49, s = 2.739, n = 9;  H0: μ = 47, H1: μ ≠ 47
t = (49 − 47)/(2.739/√9) = 2.191   with 8 d.f.
|t| = 2.19 < 3.355 (printed) → do not reject H0
```

With the printed t = 3.355 (the 1 % two-tailed value) the sample can be regarded as from a population with mean 47. (At 5 % the table value is 2.306; |t| = 2.19 would then lead to rejecting H0 — state which level you are using.)

Lesson 22 — Two-sample and Paired Tests of Means

Hyp pp. 16–31

Two independent normal samples $(n_1, \bar{x}_1, s_1^2)$ and $(n_2, \bar{x}_2, s_2^2)$; H_0: $\mu_1 = \mu_2$ ($\mu_1 - \mu_2 = 0$). $V(\bar{x}_1 - \bar{x}_2) = V(\bar{x}_1) + V(\bar{x}_2) = \sigma_1^2/n_1 + \sigma_2^2/n_2$ (variances add).

Case	Statistic	Distribution / critical value
(a) σ_1^2, σ_2^2 known	$z = (\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$	N(0, 1)
(b) $\sigma_1^2 = \sigma_2^2$ unknown, large n	$z = (\bar{x}_1 - \bar{x}_2) / [s\sqrt{(1/n_1 + 1/n_2)}]$	N(0, 1)
(b) $\sigma_1^2 = \sigma_2^2$ unknown, small n	$t = (\bar{x}_1 - \bar{x}_2) / [s\sqrt{(1/n_1 + 1/n_2)}]$	t with $n_1 + n_2 - 2$ d.f.
(c) unequal, large n	$z = (\bar{x}_1 - \bar{x}_2)/\sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	N(0, 1)
(c) unequal, small n	$t = (\bar{x}_1 - \bar{x}_2)/\sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	t_v, Welch d.f. below
Paired data	$t = \bar{d} / (s_d/\sqrt{n})$, $d_i = x_i - y_i$	t_{n-1}

Pooled variance: $s^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$

Welch d.f.: $v = (s_1^2/n_1 + s_2^2/n_2)^2 / [\,(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)\,]$

Decision rules are those of Lesson 21 with the d.f. above (right tail: reject if statistic > critical; left tail: < −critical).

TRAP Do not use the pooled s^2 unless the question says (or you have tested) that the population variances are equal — test with F = larger s^2/smaller s^2 (ANOVA note). Slide p. 21 writes $V(\bar{x}_1 - \bar{x}_2) = V(\bar{x}_1) - V(\bar{x}_2)$ by a slip; it is a plus.

Slide Examples

```
z, σ known: n1 = 50, x̄1 = 105, σ1 = 10; n2 = 60, x̄2 = 100, σ2 = 12:  z = 5/√(100/50 + 144/60) = 5/2.098 = 2.384 (p. 18)
Tribal (12, 4.5, σ² 1) vs non-tribal (15, 3.4, σ² 1.5): z = 1.1/√(1/12 + 1.5/15) = 2.57 > 1.96 → reject (p. 20)
Pooled z, large: n 40, 50; x̄ 52, 48; s² 25, 36: s = √[(39·25 + 49·36)/88] = 5.58; z = 4/(5.58√(1/40 + 1/50)) = 3.38 (p. 23)
Garment workers: n 15, x̄ 350, s 18; n 10, x̄ 345, s 21: s_p = 19.23, t = 5/(19.23√(1/15 + 1/10)) = 0.64 < t_0.05,23 = 1.714 → not rejected (p. 25)
Medicines A (12, 8, s² 9) vs B (10, 5, s² 16): t = 2.01 < t_0.025,20 = 2.086 → no significant difference (p. 25)
```

Paired t-test (pp. 28–31)

Sales of 10 companies 2014 vs 2015: d = [3, 7, 7, 3, 4, 1, 4, 2, 3, 3], $\bar{d} = 3.7$, $s_d = \sqrt{[(171 - 10 \times 3.7^2)/9]} = 1.95$, $t = 3.7/(1.95/\sqrt{10}) = 6 > t_{0.025,\,9} = 2.262$ → the two years differ significantly. Exercise (memory training, 9 students; H_1: $\mu_d > 0$, d = after − before): d = −7, 14, 5, −6, −5, 3, 1, 6, −5 → $\bar{d} = 0.667$, $t = 0.28 < t_{0.05,\,8} = 1.860$ → the training is not shown effective.

Exam Approach

1. Write H_0: $\mu_1 = \mu_2$; choose the tail from the claim.
2. Know the variances? → z. Equal and unknown? → pooled s, z (large) or t ($n_1 + n_2 - 2$ d.f.). Unequal and unknown? → Welch.
3. Compute s_p, the standard error, the statistic; compare with the critical value; conclude.

Practice — Lesson 22

(a) Actual — 2023-24 Q7(c) (22 batch p. 4). (b) Actual — 2012-13 Q8(b) (TF p. 5).

(a) Two independent samples of 8 and 7 items had the weights (ounces): I: 9, 11, 13, 11, 15, 9, 12, 14; II: 10, 12, 10, 14, 9, 8, 10. Is the difference between the sample means significant at 5 % ($t_{0.05}$ = 2.16 for 13 d.f.)? (13)

(b) 500 women shoppers in supermarket A have mean weekly food expenditure Tk 500 with SD 80; 500 women in supermarket B have Tk 440 with SD 90. Test at the 10 % level whether the average weekly food expenditures of the two populations are equal. (17)

✓ Solution

(a) Equal variances assumed (small samples) → pooled t.

```
x̄₁ = 11.75, s₁² = 4.786;  x̄₂ = 10.429, s₂² = 3.952
s² = (7 × 4.786 + 6 × 3.952)/13 = 4.401,  s = 2.098
t = (11.75 − 10.429)/(2.098 √(1/8 + 1/7)) = 1.217
H0: μ1 = μ2, H1: μ1 ≠ μ2, α = 0.05: critical |t| > 2.16;  |1.22| < 2.16 → do not reject H0
```

The difference between the two sample means is not significant at 5 %.

(b) n = 500 each (large): σ estimated by s; unequal variances ($80^2 \neq 90^2$) → z with separate variances.

```
H0: μ1 = μ2   H1: μ1 ≠ μ2   α = 0.10, critical |z| > 1.645
z = (500 − 440)/√(80²/500 + 90²/500) = 60/√(12.8 + 16.2) = 60/5.385 = 11.14
|11.14| > 1.645 → reject H0 (p ≈ 0)
```

The average weekly food expenditures of the two groups of women shoppers are not equal; supermarket A's shoppers spend more.

Lesson 23 — Tests for a Correlation Coefficient and for Proportions

Hyp pp. 32–37

1. Test of ρ = 0 (pp. 32–34)

$$H_0: \rho = 0 \text{ against } H_1: \rho \neq 0; \quad t = r\sqrt{(n-2)}/\sqrt{(1-r^2)} \sim t_{n-2}$$

Temperature vs reaction rate (7 pairs): the slide uses r = 0.995, $t = 0.995\sqrt{5}/\sqrt{(1-0.995^2)} = 22.26 > t_{0.025,\,5} = 2.571$ → reject H_0 (the exact r is 0.9986, t = 42.1, the same decision): a significant positive correlation. Voltage vs current (6 pairs, α = 0.01): r = 0.9984, $t = 35 > t_{0.005,\,4} = 4.604$ → significant.

2. One-sample Proportion Test (pp. 34–35)

$$H_0: p = p_0; \quad z = (\hat{p} - p_0) / \sqrt{[p_0(1-p_0)/n]}, \quad \hat{p} = x/n$$

Valves: $p_0 = 0.04$, 8 defective in 100: $\hat{p} = 0.08$, $z = 0.04/\sqrt{(0.04 \times 0.96/100)} = 2.04 > 1.96$ → the defect rate has changed (increased).

3. Two-sample Proportion Test (pp. 36–37)

$$H_0: p_1 = p_2; \quad z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)]}, \quad \hat{p} = (x_1 + x_2)/(n_1 + n_2)$$

LEDs: 12/200 = 0.06 and 3/150 = 0.02; pooled $\hat{p} = 15/350 = 0.04286$; SE = 0.0219; $z = 1.83 < 1.96$ → no significant difference.

Exercise answers (p. 37): (1) $\hat{p} = 10/120 = 0.0833$, $H_1: p > 0.05$: $z = 1.68$ → < 1.645, not increased; (2) 9/150 vs 4/130, $\hat{p} = 13/280$: $z = 1.16$ → no difference; (3) 4/80 vs 0.02, H_1: p > 0.02: $z = 1.92 > 1.645$ → increased; (4) 5/100 vs 2/120, 1 %: $z = 1.4 < 2.58$ → no difference.

Exam Approach

1. Proportions: $\hat{p} = x/n$; H_0: $p = p_0$ (one sample) or $p_1 = p_2$ (two samples, pooled $\hat{p}$).
2. z with the standard error under H_0; compare with 1.645 / 1.96 / 2.58 as the level and tail require.
3. Correlation: compute r (Correlation note), t with n − 2 d.f.

Practice — Lesson 23

Actual — 2023-24 Q7(b) (22 batch p. 3).

A machine puts out 16 imperfect articles in a sample of 500. After the machine is overhauled it puts out 3 imperfect articles in a batch of 100. Has the machine improved at the 1 % level of significance (z = ±2.58)? (12)

✓ Solution

```
H0: p1 = p2   H1: p1 > p2 (the old rate is higher, i.e. improvement)   α = 0.01
p̂1 = 16/500 = 0.032;  p̂2 = 3/100 = 0.03;  pooled p̂ = 19/600 = 0.03167
SE = √[0.03167 × 0.96833 × (1/500 + 1/100)] = 0.01918
z = (0.032 − 0.03)/0.01918 = 0.104
Critical: 2.33 (one-tailed) or ±2.58 (two-tailed, as printed); 0.1 < 2.33 → do not reject H0
```

There is no significant evidence at 1 % that the overhaul reduced the proportion of imperfect articles (the sample rates are 3.2 % before and 3 % after).

Lesson 24 — χ^2 Tests: Contingency Tables and Goodness of Fit

Hyp pp. 38–44 · goodness of fit: textbook

1. r × p Contingency Table (pp. 38–41)

A contingency table cross-classifies two attributes A (r categories) and B (p categories), categories mutually exclusive and exhaustive. O_{ij} = observed cell frequency, R_i, C_j = row and column totals, N = total.

$$H_0\text{: no association between A and B;} \quad E_{ij} = R_i C_j / N$$
$$\chi^2 = \Sigma\Sigma (O_{ij} - E_{ij})^2/E_{ij} = \Sigma\Sigma O_{ij}^2/E_{ij} - N \sim \chi^2_{(r-1)(p-1)}$$
$$\text{Reject } H_0 \text{ if } \chi^2 \geq \chi^2_{\alpha,\,(r-1)(p-1)} \text{ (one-tailed, upper tail)}$$

Education vs job performance (pp. 40–41) — also 2023-24 Q8(a)

Cell (i, j)	O	E	$(O - E)^2/E$
(1,1)	10	15	1.67
(1,2)	40	30	3.33
(1,3)	10	15	1.67
(2,1)	30	20	5
(2,2)	30	40	2.5
(2,3)	20	20	0
(3,1)	10	15	1.67
(3,2)	30	30	0
(3,3)	20	15	1.67
Total	200	200	$17.5 = \chi^2$

Expected: $E_{11} = 60 \times 50/200 = 15$, $E_{21} = 80 \times 50/200 = 20$, $E_{22} = 80 \times 100/200 = 40$, $E_{32} = 60 \times 100/200 = 30$ (the slide's list gives 30 for E_{21}; the table uses the correct 20). $\chi^2 = 17.5$ with $(3 - 1)(3 - 1) = 4$ d.f.; $\chi^2_{0.05,\,4} = 9.49$; $17.5 \geq 9.49 \rightarrow$ reject H_0: job performance is associated with educational attainment ($p = 0.0015$).

2. 2 × 2 Table (pp. 42–44)

	B_1	B_2	Total
A_1	a	b	a + b
A_2	c	d	c + d
Total	a + c	b + d	N

$$\chi^2 = N(|ad - bc| - N/2)^2 / [(a + b)(a + c)(c + d)(b + d)] \sim \chi^2_1 \quad \text{(with Yates' correction)}$$
$$\text{without correction: } \chi^2 = N(ad - bc)^2 / [(a + b)(a + c)(c + d)(b + d)]$$

Practice problems of the lecture (pp. 43–44) — answers (Yates-corrected): smoking vs lung disease [36, 14; 18, 32]: $\chi^2 = 11.63 > 3.84 \rightarrow$ dependent; online course [42, 18; 30, 30]: $\chi^2 = 4.2$ (< 3.84, no association at 5 %); job vs gender [40, 20; 25, 35]: $\chi^2 = 6.58 > 3.84 \rightarrow$ job type depends on gender. Exercises (p. 42): gender × major (141 students) and immunisation × mother's education (500 children) are r × p tables solved as above.

3. Goodness of Fit (textbook; pp. 5–6 state the statistic)

$$\chi^2 = \Sigma\,(O_i - E_i)^2/E_i \sim \chi^2_{k-1-m}, \quad E_i = N\,p_i$$

k = number of classes, m = number of parameters estimated from the data; use classes with $E_i \geq 5$

4. What "Goodness of Fit" Means

It tests whether a set of observed frequencies is consistent with a theoretical distribution (a coin is fair, births are equally likely male/female, counts are Poisson). H_0: the data follow the specified distribution; a large χ^2 means the observed frequencies are far from the expected ones.

Exam Approach

1. Contingency: row/column totals, $E_{ij} = R_iC_j/N$, χ^2 by the table, d.f. = $(r-1)(p-1)$, compare with $\chi^2_{\alpha,\,\text{d.f.}}$.
2. 2 × 2: use the ad − bc formula (state whether you apply Yates' correction).
3. Goodness of fit: $E_i = N\,p_i$ (e.g. binomial(5, 1/2) × N), $\chi^2 = \Sigma(O-E)^2/E$, d.f. = k − 1 − m.
4. Conclude in words (association exists / births equally probable / coin unbiased).

Practice — Lesson 24

(a) Actual — 2022-23 Q8(a) (TF p. 50). (b) Actual — 2023-24 Q8(a) (22 batch p. 4). (c) Actual — 2011-12 Q8(b) (TF p. 3). (d) Actual — 2012-13 Q7(b) (TF p. 5). (e) Actual — 2013-14 Q8(b) (TF p. 8).

(a) A drug was given to 500 of 800 people to test its efficacy against typhoid. Typhoid: drug 200, no drug 280; no typhoid: drug 300, no drug 20. Can it be concluded that the drug is effective in preventing typhoid at 5 % ($\chi^2_{0.05,\,1} = 3.84$)? (10)

(b) A random sample of 200 employees (education × job performance table, TF 22 batch p. 4) — is there a relationship? Test at 5 % ($\chi^2_{0.05,\,4} = 9.49$). (15)

(c) 5 coins are tossed 3200 times; the number of heads 0, 1, 2, 3, 4, 5 occurs 80, 570, 1100, 900, 500, 50 times. Test at 5 % that the coins are unbiased ($\chi^2_{0.05,\,5} = 11.07$). (18)

(d) A survey of 640 families with 5 children each: boys 5, 4, 3, 2, 1, 0 → 28, 112, 220, 176, 80, 24 families. Is the result consistent with the hypothesis that male and female births are equally probable? 5 %, $\chi^2_{0.05,\,5} = 11.07$. (17)

(e) A survey of 320 families with 5 children: boys 5, 4, 3, 2, 1, 0 → 18, 56, 110, 88, 40, 8 families. Same question. (15)

✓ Solution

(a) H0: drug and typhoid are independent (the drug is not effective); H1: associated. N = 800.

```
E(drug, typhoid) = 500 × 480/800 = 300;  E(drug, no) = 200;  E(no drug, typhoid) = 180;  E(no, no) = 120
χ² = (200 − 300)²/300 + (300 − 200)²/200 + (280 − 180)²/180 + (20 − 120)²/120
   = 33.33 + 50 + 55.56 + 83.33 = 222.22   (formula N(ad − bc)²/(…) = 222.22)
with Yates' correction: N(|ad − bc| − N/2)²/(…) = 220.01
d.f. = 1, critical 3.84: 220 > 3.84 → reject H0
```

The drug is associated with typhoid: 40 % of the treated got typhoid against 93 % of the untreated, so the drug is effective in preventing typhoid.

(b) Worked in §1: $\chi^2 = 17.51 > 9.49$ with 4 d.f. → there is a relationship between educational attainment and job performance. (Excellent performers are over-represented among college graduates, 40 observed against 30 expected.)

(c) H0: the coins are unbiased, so heads ~ Bin(5, ½): probabilities 1, 5, 10, 10, 5, 1 over 32; E = 3200 × these = 100, 500, 1000, 1000, 500, 100.

Heads	O	E	$(O-E)^2/E$
0	80	100	4
1	570	500	9.8
2	1100	1000	10
3	900	1000	10
4	500	500	0
5	50	100	25
Total	3200	3200	$\chi^2 = 58.8$

d.f. = 6 − 1 = 5, $\chi^2_{0.05,\,5} = 11.07$; $\chi^2 = 58.8 > 11.07$ → reject H0: the coins cannot be regarded as unbiased (too few 0-head and 5-head throws, too many 1- and 3-head).

(d) H0: P(boy) = ½, so boys among 5 children ~ Bin(5, ½); E = 640 × (1, 5, 10, 10, 5, 1)/32 = 20, 100, 200, 200, 100, 20 for 5, 4, …, 0 boys.

```
χ² = (28 − 20)²/20 + (112 − 100)²/100 + (220 − 200)²/200 + (176 − 200)²/200 + (80 − 100)²/100 +
(24 − 20)²/20
   = 3.2 + 1.44 + 2 + 2.88 + 4 + 0.8 = 14.32
d.f. = 5;  14.32 > 11.07 → reject H0
```

The data are not consistent with equally probable male and female births (more boys than girls: 1.18 boys per girl).

(e) E = 320 × (1, 5, 10, 10, 5, 1)/32 = 10, 50, 100, 100, 50, 10.

```
χ² = (18 − 10)²/10 + (56 − 50)²/50 + (110 − 100)²/100 + (88 − 100)²/100 + (40 − 50)²/50 + (8 −
10)²/10
   = 6.4 + 0.72 + 1 + 1.44 + 2 + 0.4 = 11.96
d.f. = 5;  11.96 > 11.07 → reject H0
```

Again the hypothesis of equally probable male and female births is rejected at 5 %.

Quick Notes — Tests of Hypothesis

Revision points for Lessons 20–24.

L20 · Concepts

- H0 has '='; H1 sets the tail. α = P(Type I) = P(reject H0 | true); β = P(Type II).
- p-value ≤ α → reject. Two-tailed p = 2P(Z ≥ |z|).
- DNA: Type I = excluding the real father (≈ 0.001); Type II = declaring a non-father the father (0).

L21 · One sample

- z = (x̄ − μ0)/(σ/√n) (σ known or n ≥ 30); t = (x̄ − μ0)/(s/√n), n − 1 d.f., s with divisor n − 1.
- Critical z: 1.645 (5 % one-tail), 1.96 (5 % two), 2.33 (1 % one), 2.58 (1 % two). CI x̄ ± z σ/√n.
- Results: mice z = −2.76, reject; deaths z = 3.67, reject; GPA z = 7.23, CI (2.66, 2.94); cereal t = 4.38; sales t = 4.29; waiting t = −2.33; delivery z = −4.67; heights t = 2.02 (no); tins t = −3.59 (reject).

L22 · Two samples

- Variances add: σ1²/n1 + σ2²/n2. Pooled s² = [(n1−1)s1² + (n2−1)s2²]/(n1+n2−2), t with n1 + n2 − 2 d.f.
- Paired t = d̄/(s_d/√n). Samples I/II: t = 1.22 (no difference); supermarkets z = 11.14 (differ).

L23 · r and proportions

- t = r√(n−2)/√(1−r²) with n − 2 d.f.; z = (p̂ − p0)/√(p0q0/n); two-sample pooled p̂.
- Machine overhaul: z = 0.1 (not significant).

L24 · χ²

- E_ij = R_i C_j/N; χ² = Σ(O − E)²/E; d.f. (r − 1)(p − 1); 2 × 2: N(ad − bc)²/(…), Yates uses |ad − bc| − N/2.
- Typhoid χ² = 222.22/220.01; 3 × 3 job table 17.51 (reject); coins 58.8; families 14.32 and 11.96 (all reject, 11.07).